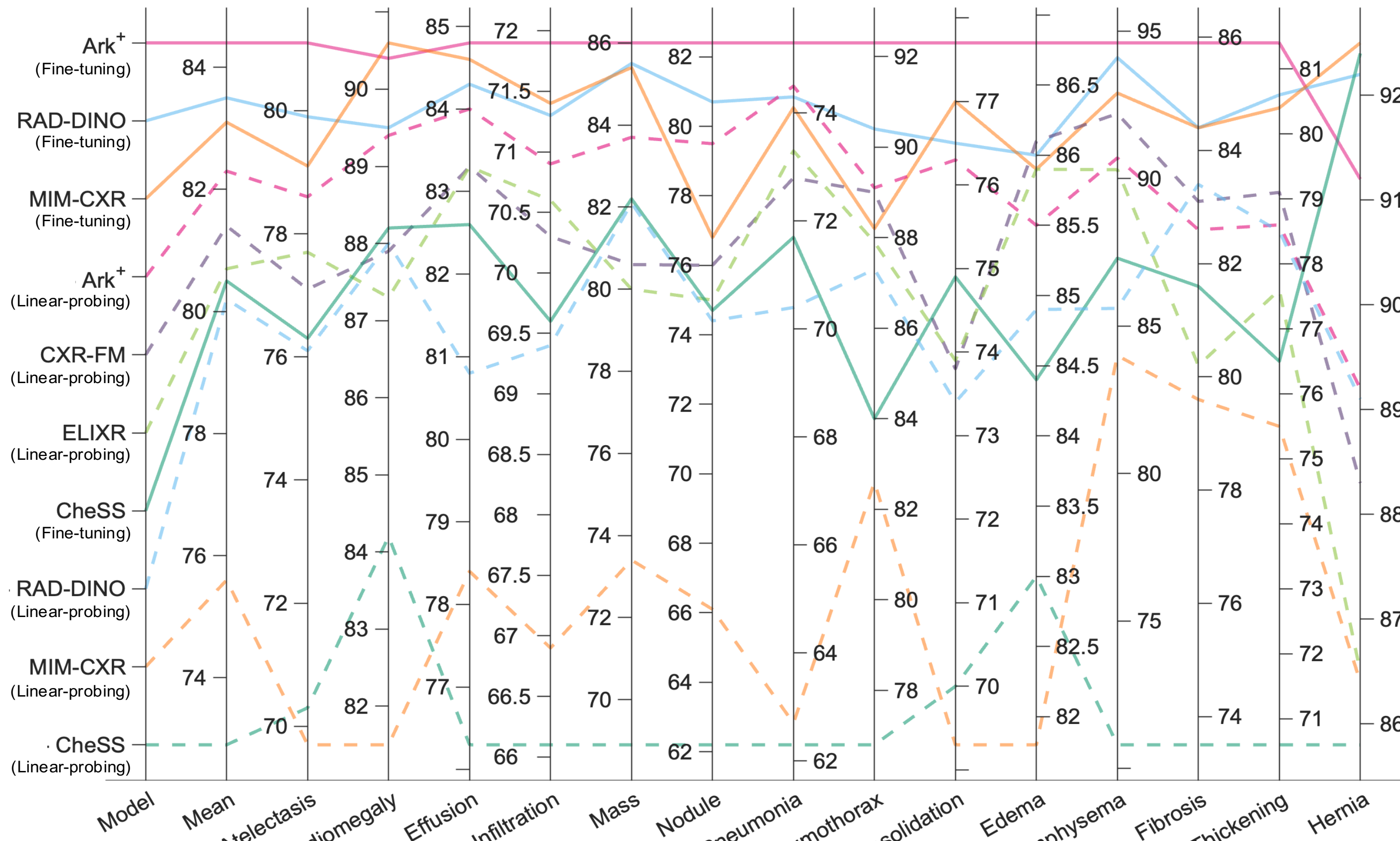


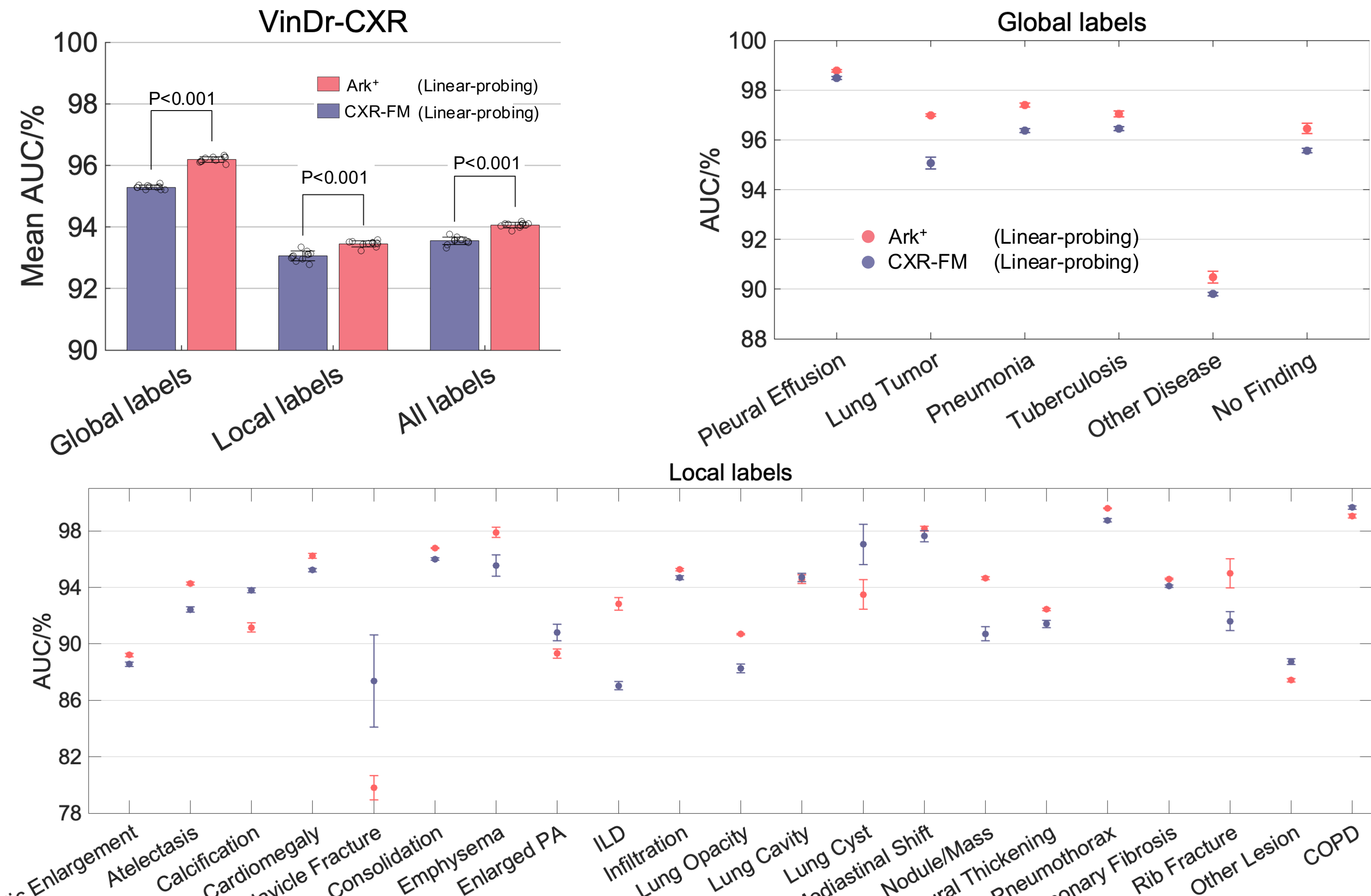
Result 1: Diagnose common thoracic diseases (ChestXray14)

- Ark⁺ is a superior open foundation model that surpasses state-of-the-art counterparts under both fine-tuning and linear-probing settings, demonstrating strong generalizability.



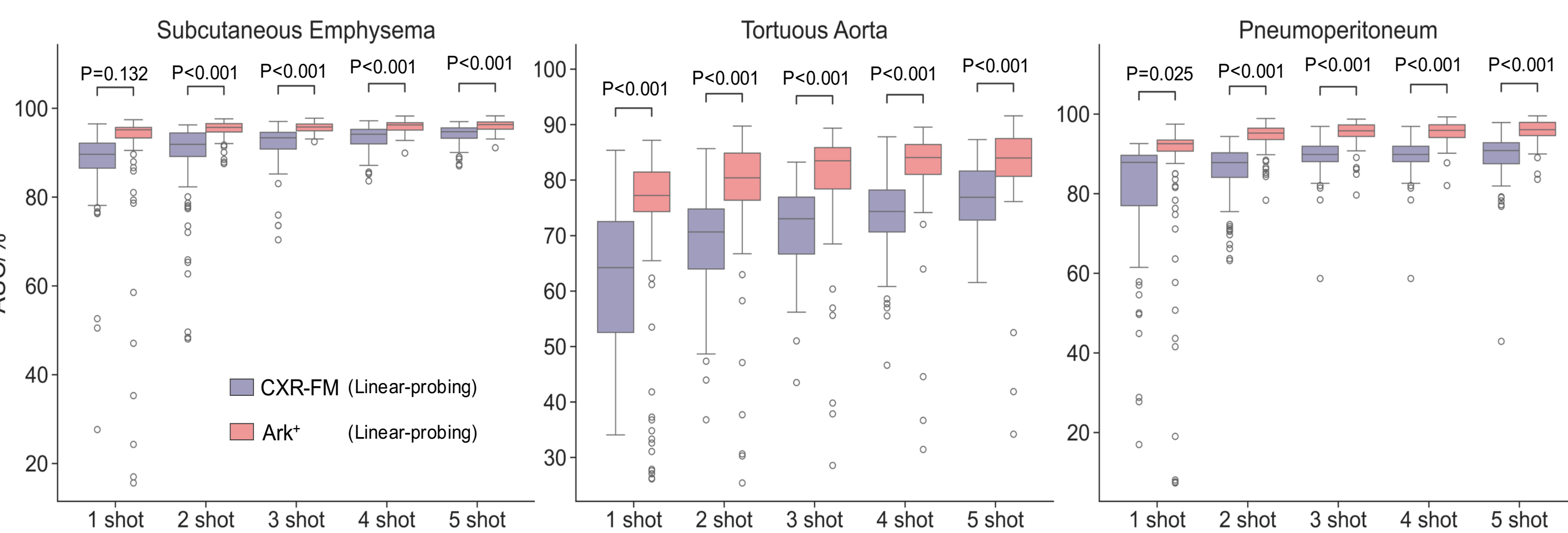
Result 2: Adapt to evolving diagnostic needs (VinDr-CXR)

- Ark⁺, pretrained on the six global labels of VinDr-CXR, rapidly adapt to the 22 local labels without requiring extensive retraining, demonstrating strong adaptability to evolving diagnostic needs.

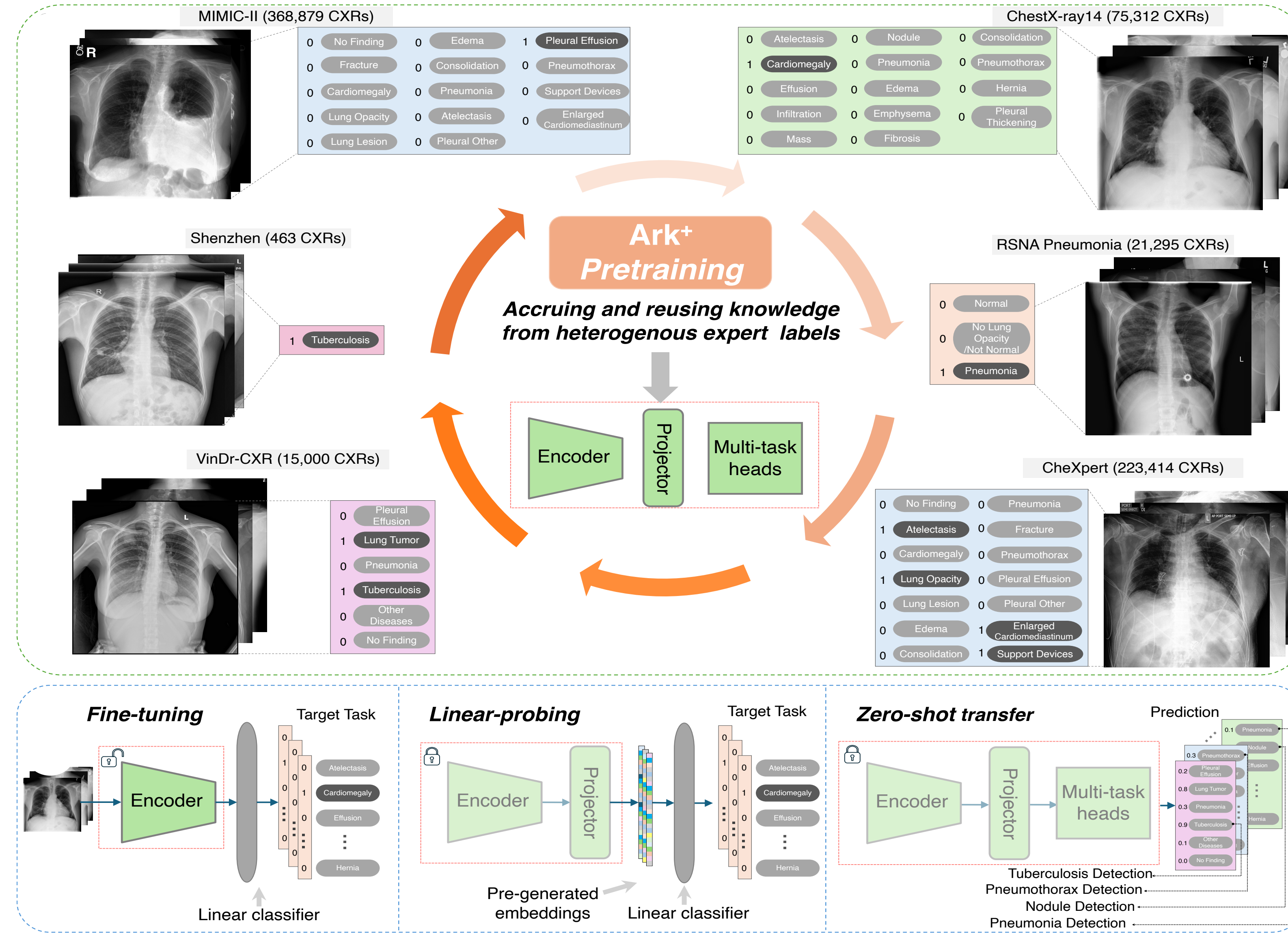


Result 3: Learn rare conditions from a few samples (CXR-LT)

- Ark⁺ can diagnose rare disorders from only a few labeled samples, demonstrating strong potential for clinical use in data-scarce settings.

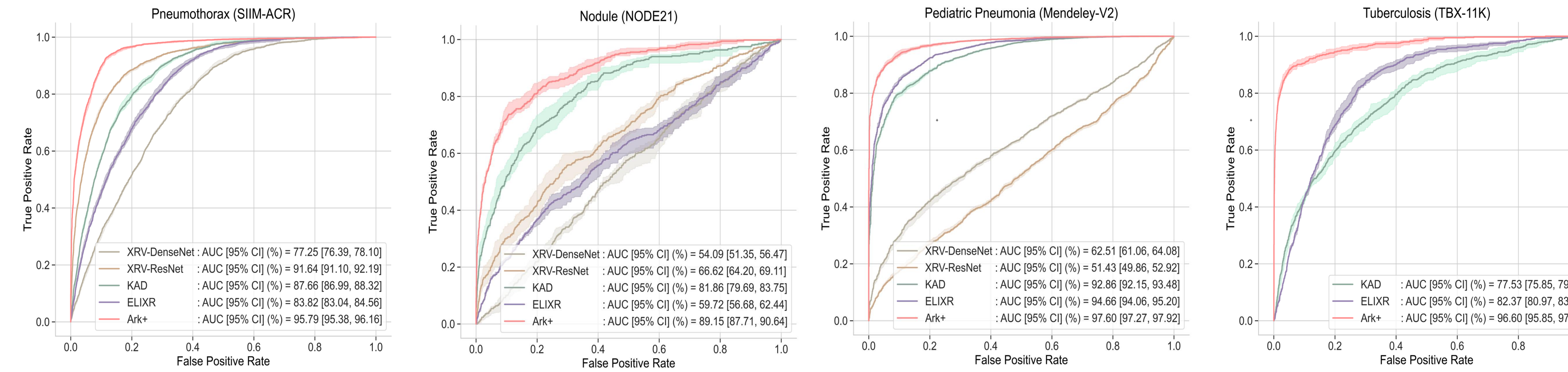


Ark⁺, a fully open AI foundation model using a multitude of datasets with heterogeneous labels, demonstrates exceptional capabilities in diagnosing common, rare and novel thoracic diseases



Result 5: Transfer to new sites without training

- Ark⁺ can transfer to unseen domains and diverse clinical contexts without additional training (zero-shot transfer).



Result 6: Tolerate sex-related bias

- Ark⁺ tolerates extreme data imbalance and reduces sex-related bias, thereby improving fairness and diagnostic accuracy.

AUC(%)	CheSS	MIM-CXR	CXR-FM	Ark ⁺
Dz ⁺ Test data	Training data	Training data	Training data	Training data
	Female	Female	Female	Female
EC	Female 63.92±1.4	61.82±1.3	69.53±1.4	70.41±1.4
Male 66.11±1.6	64.79±1.4	68.04±1.5	71.63±1.5	69.99±1.5
CM	Female 79.37±0.8	72.99±1.1	72.28±1.3	84.15±0.7
Male 80.00±0.8	81.47±0.9	73.84±1.0	85.11±0.8	85.59±0.9
LO	Female 67.74±0.7	67.16±0.7	68.96±0.7	68.11±0.7
Male 72.27±0.5	73.32±0.5	73.44±0.8	74.10±0.4	77.24±0.4
LL	Female 63.03±1.4	62.79±1.8	66.71±1.9	65.57±2.2
Male 64.81±1.7	66.69±1.6	67.50±1.3	69.36±1.3	77.29±1.2
ED	Female 78.60±0.7	78.32±0.8	78.33±0.7	77.52±0.8
Male 81.95±0.6	82.29±0.6	81.86±0.6	82.43±0.6	86.04±0.4
CS	Female 67.72±1.3	67.23±1.5	67.27±1.5	66.93±1.5
Male 67.45±1.1	68.18±1.3	66.62±1.6	67.01±1.5	75.27±1.0
PN	Female 62.05±2.2	62.25±2.7	63.85±1.5	64.24±1.4
Male 65.48±2.2	66.81±1.7	65.13±2.0	67.50±2.2	78.38±1.7
AT	Female 63.89±0.8	63.04±0.7	64.77±1.1	64.36±1.2
Male 67.05±0.9	67.81±0.8	67.78±0.8	68.74±0.5	72.51±0.5
PX	Female 72.95±1.1	71.36±1.2	77.69±1.2	76.14±1.3
Male 67.10±2.0	68.46±1.5	74.58±1.4	76.22±1.7	83.02±0.9
PE	Female 79.73±0.6	79.04±0.5	81.15±0.5	80.02±0.6
Male 81.64±0.6	82.38±0.6	83.61±0.6	84.26±0.5	89.53±0.5
PO	Female 70.45±2.8	67.89±3.5	73.05±2.6	71.47±3.0
Male 65.42±2.0	69.50±2.9	69.58±3.0	72.08±2.6	77.12±2.2
FR	Female 70.57±1.5	65.70±2.0	73.88±1.3	70.29±1.7
Male 61.24±1.6	66.04±1.5	66.69±1.2	72.68±1.3	76.28±1.4

Result 7: Respond to novel diseases (COVID-19)

- Ark⁺ exhibits adaptability and extensibility in handling previously unseen diseases and novel clinical conditions, such as potential future pandemics.

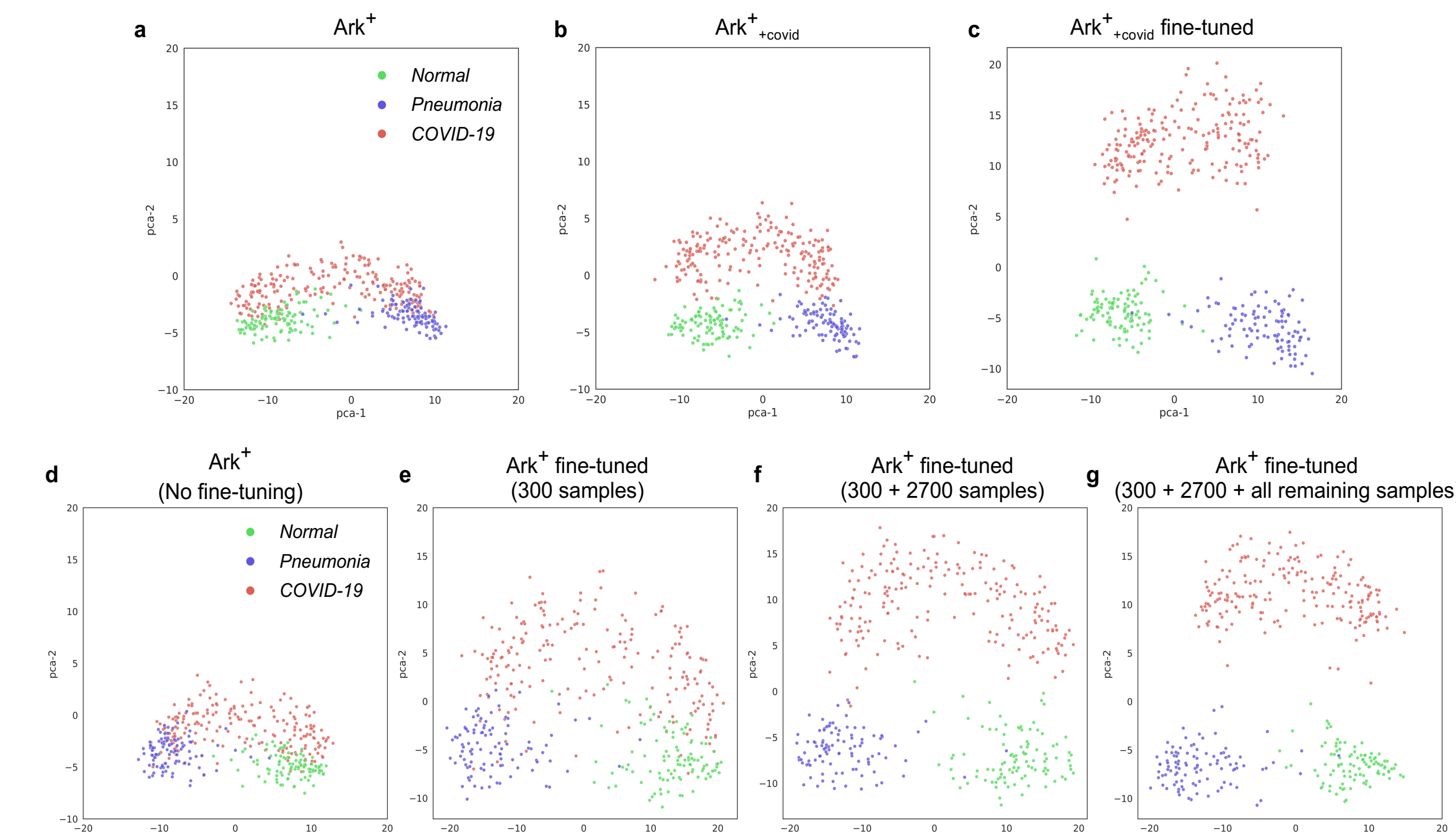
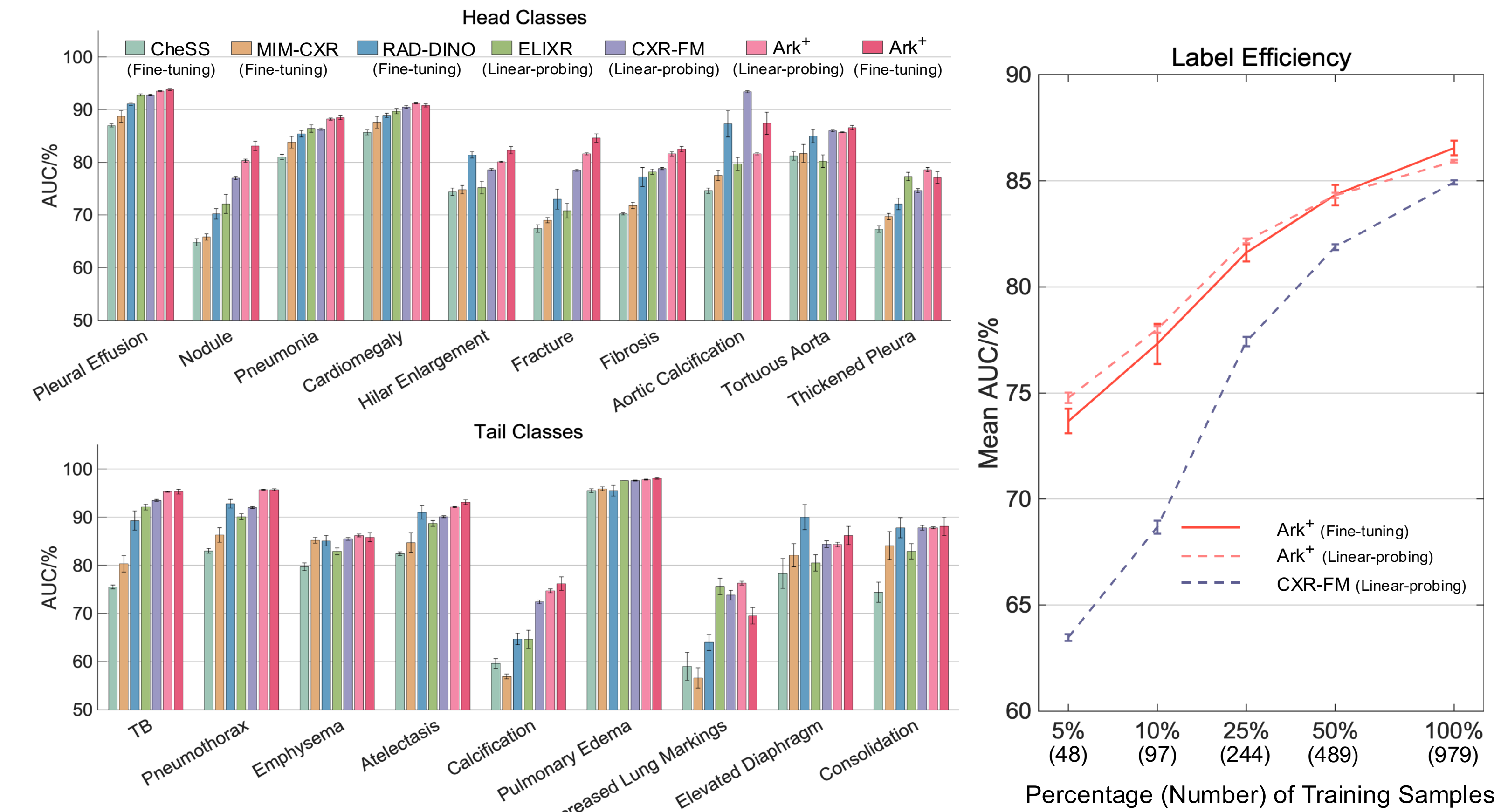


Illustration of how the embeddings for COVID-19, Pneumonia and Normal evolve in t-SNE

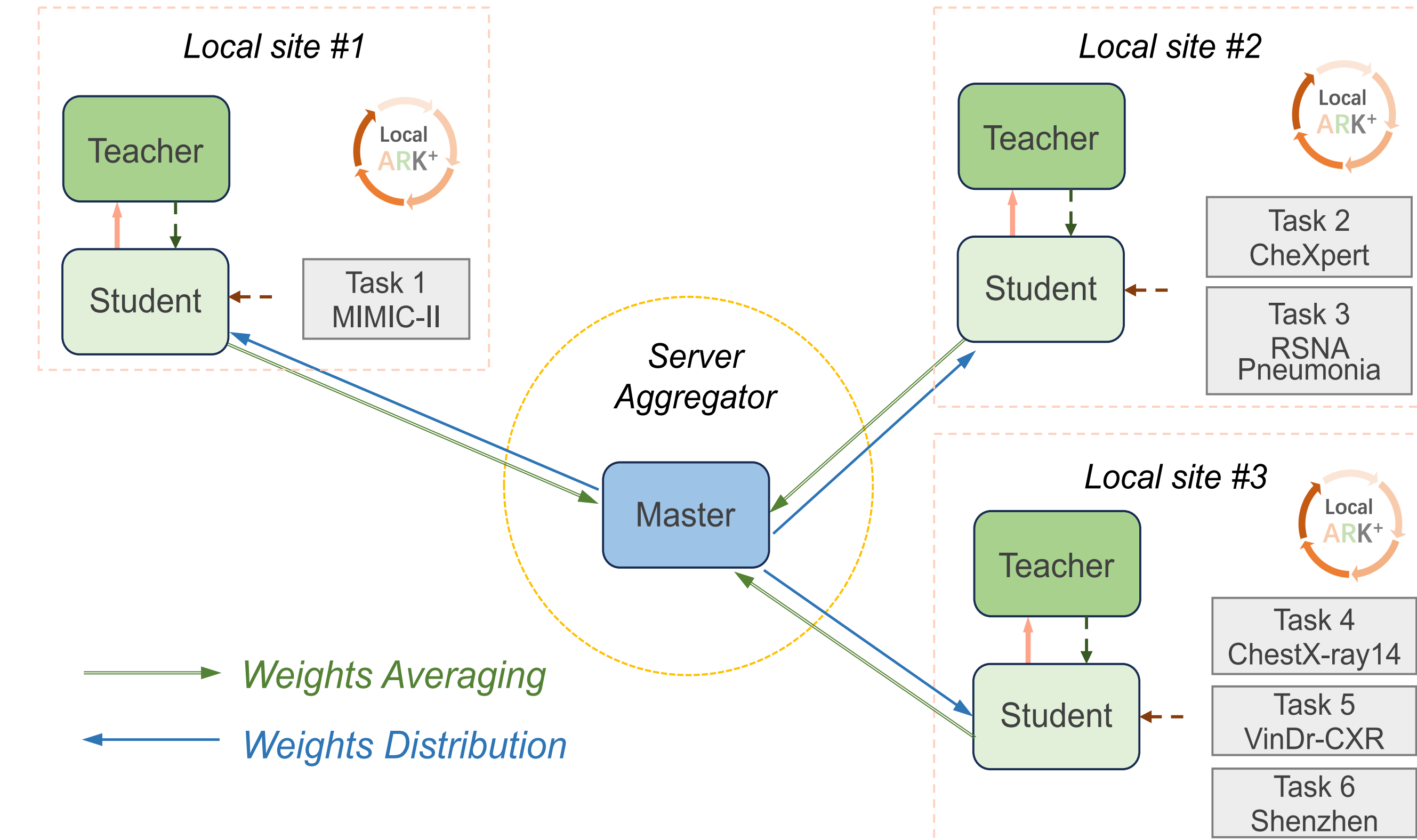
Result 4: Handle long-tailed thoracic diseases (ChestDR)

- Ark⁺ demonstrates strong robustness and generalizability under long-tailed data distributions.



Result 8: Protect privacy and distribute pretraining

- Federated Ark⁺ delivers competitive performance in scenarios that require privacy preservation and distributed computation across multiple sites.



Illustrating a federated learning scenario by distributing Ark⁺'s training across three local sites

Pretraining task	Metrics	3-site	Isolated (local site) training	Federated training	Centralized training
MIMIC-II	AUC	Site #1	79.67	79.41	80.37
CheXpert	AUC	Site #2	87.60	88.53	89.60
RSNA Pneumonia	Accuracy	Site #2	75.45	75.78	75.56
ChestX-ray14	AUC	Site #3	81.61	83.16	84.15
VinDr-CXR	AUC	Site #3	94.63	96.17	96.63
Shenzhen	AUC	Site #3	98.66	99.29	99.29
Average performance (%)			86.27	87.06	87.60

Performance comparison on COVID-19 diagnostic task

Model	ACC(%)	meanAUC(%)	meanMCC(%)	meanAP(%)	meanF1(%)
fine-tuning*					
MIM-CXR	95.88±1.06	99.11±0.47	93.88±2.02	97.10±1.65	95.67±1.37
RAD-DINO	97.05±0.27	99.64±0.14	95.58±0.81	99.08±0.30	96.81±0.56
Ark ⁺	98.83±0.26	99.87±0.14	97.35±0.32	99.60±0.19	98.59±0.47
Ark ⁺ +covid	99.08±0.26	99.96±0.01	97.72±0.17	99.89±0.03	98.33±0.03
linear-probing					
CXR-FM	98.60±0.46	99.80±0.04	97.07±0.32	99.45±0.09	98.14±0.27
MIM-CXR	95.50±0.16	99.16±0.14	91.87±0.49	97.60±0.26	94.57±0.38
RAD-DINO	95.85±0.14	99.07±0.09	92.28±0.33	98.08±0.16	94.58±0.25
Ark ⁺	95.87±0.20	99.09±0.10	93.04±0.28	97.80±0.33	95.49±0.19
Ark ⁺ +covid	99.10±0.12	99.90±0.06	97.92±0.21	99.61±0.36	98.97±0.16